

AWS VPC Challenge

Designing a Scalable, Secure Multi-VPC Architecture

Overview & Objectives

This runbook provides a step-by-step implementation guide for building a scalable, secure multi-VPC network architecture on AWS using AWS Transit Gateway and VPC Endpoints (Gateway and Interface types). It is intended for use during an on-premises-to-cloud migration where multiple departmental and application VPCs must communicate with centralized services while minimizing exposure to the public internet.

Objectives

- Design and deploy a scalable network architecture using AWS Transit Gateway to simplify connectivity between multiple VPCs.
- Configure VPC Endpoints to securely access AWS services without relying on internet gateways or NAT gateways.
- Maintain data privacy and minimize exposure to external threats across all connected VPCs.
- Produce a documented, repeatable process the team can use for future VPC onboarding.

Prerequisites

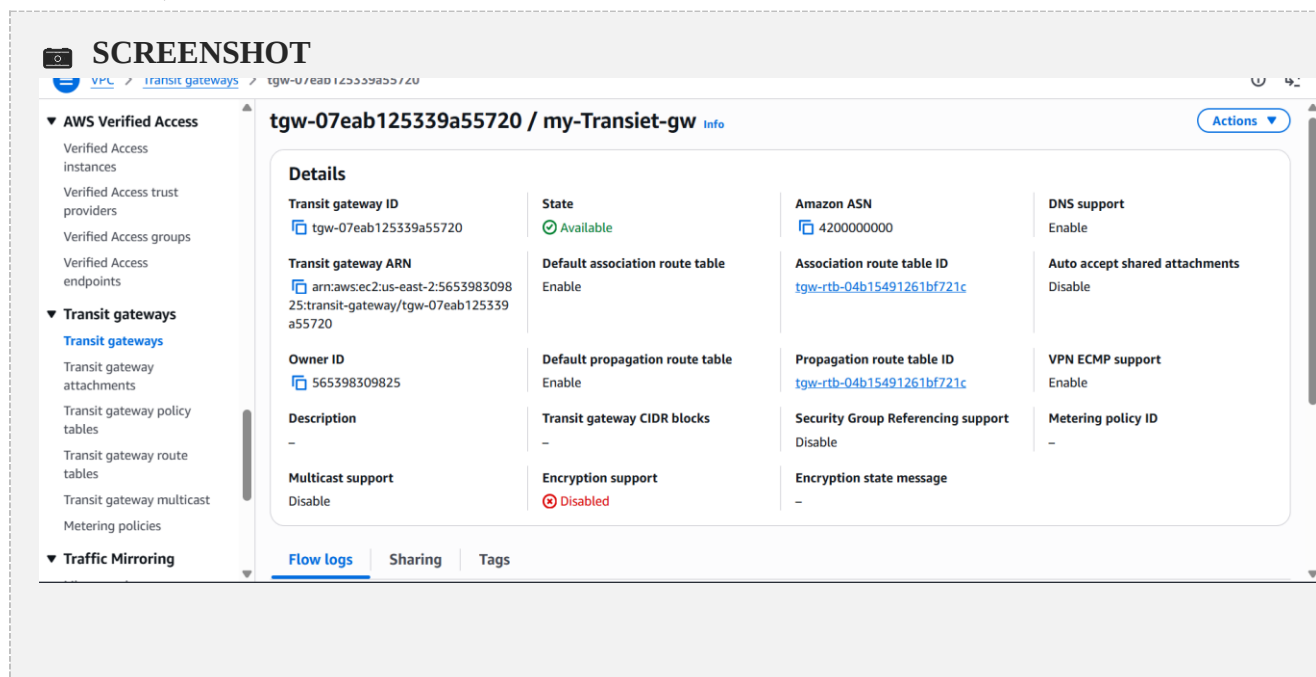
- An AWS account with administrative or network-admin IAM permissions.
- At least two existing VPCs (e.g., vpc-shared-services, vpc-app-dept-a) in the same AWS Region, with non-overlapping CIDR blocks.
- A defined CIDR / IP addressing plan to avoid overlaps across VPCs and on-premises networks.
- Access to the AWS Management Console (for screenshots) and/or AWS CLI v2.

Design & Deploy AWS Transit Gateway

Transit Gateway acts as a central hub that simplifies routing between multiple VPCs and on-premises networks, replacing a complex mesh of VPC peering connections with a single, scalable connection point.

1.1. Create the Transit Gateway

1. Sign in to the AWS Management Console and navigate to VPC > Transit Gateways.
2. Choose Create Transit Gateway.
3. Enter a Name tag : my-Transient-gw, and an optional description.
4. Leave Amazon side ASN at the default (64512) unless your organization requires a specific ASN for BGP.
5. Under Default route table association and Default route table propagation, leave both Enabled for a simple hub-and-spoke start (these will be refined into segmented route tables in step 1.3).
6. Enable DNS support and leave Multicast support disabled unless required.
7. Choose Create Transit Gateway and wait for the state to change from pending to available (typically 1–2 minutes).

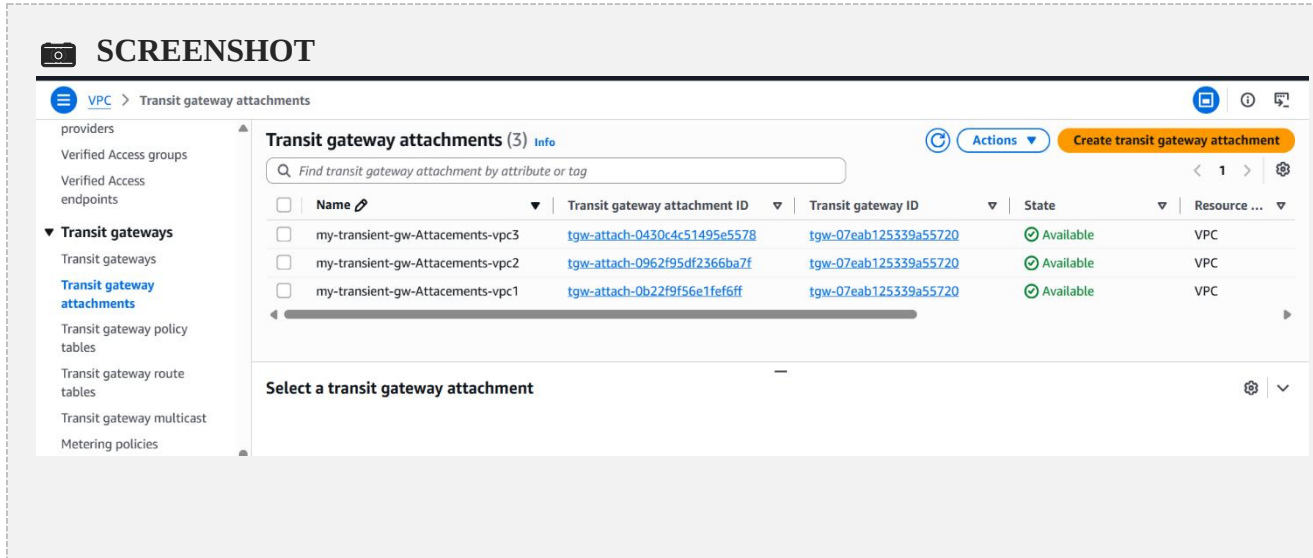


1.2. Attach VPCs to the Transit Gateway

Each VPC that needs to participate in the shared network must be attached to the Transit Gateway through a VPC attachment.

8. In the VPC console, go to Transit Gateway Attachments > Create Transit Gateway Attachment.
9. Select the Transit Gateway created in step 1.1., my-Transient-gw.
10. Set Attachment type to VPC.

11. Choose the target VPC (e.g., vpc-shared-services) and select one subnet per Availability Zone you want the attachment to use.
12. Leave DNS support enabled so private DNS resolution works across attached VPCs.
13. Choose Create attachment and repeat this process for every VPC that needs connectivity (e.g., vpc-app-dept-a, vpc-app-dept-b).
14. Confirm each attachment reaches the Available state before proceeding.



1.3. Configure Transit Gateway Route Tables

Route tables on the Transit Gateway control which attachments can reach which other attachments. Using segmented route tables (rather than the single default table) lets you enforce isolation between departments while still allowing shared access to central services.

15. Go to Transit Gateway Route Tables > Create Transit Gateway Route Table.
16. Create one route table per network segment, e.g., rtb-tgw-shared-services and rtb-tgw-app-vpcs.
17. For each Transit Gateway attachment, go to the Associations tab of the appropriate route table and choose Create association to link the attachment.
18. Under Propagations, associate each attachment so its VPC CIDR is automatically propagated into the relevant route table(s).
19. For attachments that should only reach shared services (not each other), create static routes or limit propagation so cross-department traffic is not automatically permitted.
20. Disable the default association and propagation settings from step 1.1 once the custom route tables are validated, to prevent unintended full-mesh connectivity.

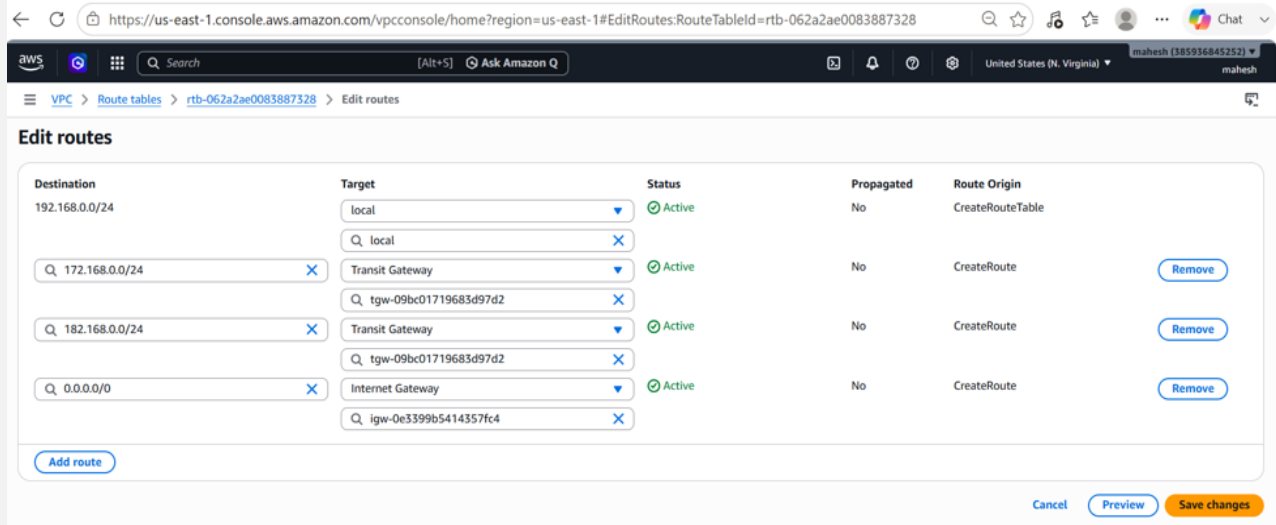
1.4. Update VPC Route Tables

Each attached VPC's own subnet route tables must point traffic destined for other VPCs (or shared services) toward the Transit Gateway.

21. In the VPC console, open Route Tables and select the route table associated with the relevant subnet(s).
22. Choose Edit routes > Add route.

23. Set Destination to the CIDR block of the remote VPC or a summarized range covering all spoke VPCs (e.g., 10.0.0.0/8).
24. Set Target to the Transit Gateway (tgw-corp-hub).
25. Save the route and repeat for every subnet route table in every attached VPC that needs cross-VPC connectivity.

SCREENSHOT



1.5. Validate Transit Gateway Connectivity

26. Launch or use an existing test EC2 instance in each attached VPC.
27. From an instance in vpc-app-dept-a, run a reachability test (e.g., ping or curl) to a private IP address of an instance in vpc-shared-services.
28. Confirm security groups and network ACLs on both sides allow the test traffic.
29. Use VPC Reachability Analyzer (VPC console > Reachability Analyzer) for an automated, non-intrusive path check between the two ENIs.

SCREENSHOT

Connection From Vpc1 to Vpc2 And Vpc3

```

DELL@KHADEER MINGW64 ~/Downloads
$ ssh -i Ohiokey.pem ec2-user@3.16.23.22
The authenticity of host '3.16.23.22 (3.16.23.22)' can't be established.
ED25519 key fingerprint is: SHA256:m1dx6b7H14Dk6uK5rP5W0uBN595+NTvttPeE9WmtEJQ
This host key is known by the following other names/addresses:
  ~/.ssh/known_hosts:36: 18.218.5.238
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '3.16.23.22' (ED25519) to the list of known hosts.
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html

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Last login: Sat Jul 18 14:01:38 2026 from 103.203.172.230

```

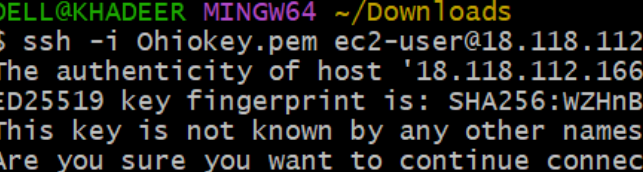
```

[ec2-user@ip-10-0-0-82 ~]$ ping 18.118.112.166
PING 18.118.112.166 (18.118.112.166) 56(84) bytes of data.
64 bytes from 18.118.112.166: icmp_seq=1 ttl=126 time=0.245 ms
64 bytes from 18.118.112.166: icmp_seq=2 ttl=126 time=0.237 ms
64 bytes from 18.118.112.166: icmp_seq=3 ttl=126 time=0.260 ms
64 bytes from 18.118.112.166: icmp_seq=4 ttl=126 time=0.222 ms
64 bytes from 18.118.112.166: icmp_seq=5 ttl=126 time=0.284 ms
64 bytes from 18.118.112.166: icmp_seq=6 ttl=126 time=0.249 ms
64 bytes from 18.118.112.166: icmp_seq=7 ttl=126 time=0.242 ms
^C
--- 18.118.112.166 ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6215ms
rtt min/avg/max/mdev = 0.222/0.248/0.284/0.018 ms
[ec2-user@ip-10-0-0-82 ~]$ ping 3.16.213.163
PING 3.16.213.163 (3.16.213.163) 56(84) bytes of data.
64 bytes from 3.16.213.163: icmp_seq=1 ttl=126 time=0.590 ms
64 bytes from 3.16.213.163: icmp_seq=2 ttl=126 time=0.466 ms
64 bytes from 3.16.213.163: icmp_seq=3 ttl=126 time=0.426 ms
64 bytes from 3.16.213.163: icmp_seq=4 ttl=126 time=0.462 ms
64 bytes from 3.16.213.163: icmp_seq=5 ttl=126 time=0.430 ms
64 bytes from 3.16.213.163: icmp_seq=6 ttl=126 time=0.453 ms
64 bytes from 3.16.213.163: icmp_seq=7 ttl=126 time=0.456 ms
64 bytes from 3.16.213.163: icmp_seq=8 ttl=126 time=0.457 ms
^C
--- 3.16.213.163 ping statistics ---
8 packets transmitted, 8 received, 0% packet loss, time 7305ms
rtt min/avg/max/mdev = 0.426/0.467/0.590/0.048 ms
[ec2-user@ip-10-0-0-82 ~]$ |

```

Connection From Vpc2 to Vpc3 And Vpc1

```
DELL@KHADEER MINGW64 ~/Downloads  
$ ssh -i Ohiokey.pem ec2-user@18.118.112.166  
The authenticity of host '18.118.112.166 (18.118.112.166)' can't be established.  
ED25519 key fingerprint is: SHA256:WZHnBkjbrbLla5rz1lKi4pQaeBeacgT/+eTb8cc0vJs  
This key is not known by any other names.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added '18.118.112.166' (ED25519) to the list of known hosts.  
** WARNING: connection is not using a post-quantum key exchange algorithm.  
** This session may be vulnerable to "store now, decrypt later" attacks.  
** The server may need to be upgraded. See https://openssh.com/pq.html
```



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Amazon Linux 2023

<https://aws.amazon.com/linux/amazon-linux-2023>

```
[ec2-user@ip-172-168-0-81 ~]$ ping 3.16.23.22
PING 3.16.23.22 (3.16.23.22) 56(84) bytes of data.
64 bytes from 3.16.23.22: icmp_seq=1 ttl=126 time=0.257 ms
64 bytes from 3.16.23.22: icmp_seq=2 ttl=126 time=0.247 ms
64 bytes from 3.16.23.22: icmp_seq=3 ttl=126 time=0.244 ms
64 bytes from 3.16.23.22: icmp_seq=4 ttl=126 time=0.259 ms
64 bytes from 3.16.23.22: icmp_seq=5 ttl=126 time=0.255 ms
64 bytes from 3.16.23.22: icmp_seq=6 ttl=126 time=0.233 ms
64 bytes from 3.16.23.22: icmp_seq=7 ttl=126 time=0.278 ms
64 bytes from 3.16.23.22: icmp_seq=8 ttl=126 time=0.259 ms
^C
--- 3.16.23.22 ping statistics ---
8 packets transmitted, 8 received, 0% packet loss, time 7248ms
rtt min/avg/max/mdev = 0.233/0.254/0.278/0.012 ms
[ec2-user@ip-172-168-0-81 ~]$ ping @3.16.213.163
ping: @3.16.213.163: Name or service not known
[ec2-user@ip-172-168-0-81 ~]$ ping 3.16.213.163
PING 3.16.213.163 (3.16.213.163) 56(84) bytes of data.
64 bytes from 3.16.213.163: icmp_seq=1 ttl=126 time=0.450 ms
64 bytes from 3.16.213.163: icmp_seq=2 ttl=126 time=0.440 ms
64 bytes from 3.16.213.163: icmp_seq=3 ttl=126 time=0.414 ms
64 bytes from 3.16.213.163: icmp_seq=4 ttl=126 time=0.434 ms
64 bytes from 3.16.213.163: icmp_seq=5 ttl=126 time=0.475 ms
64 bytes from 3.16.213.163: icmp_seq=6 ttl=126 time=0.437 ms
64 bytes from 3.16.213.163: icmp_seq=7 ttl=126 time=0.451 ms
^C
--- 3.16.213.163 ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6239ms
rtt min/avg/max/mdev = 0.414/0.443/0.475/0.017 ms
[ec2-user@ip-172-168-0-81 ~]$
```

Connection From Vpc3 to Vpc2 And Vpc1


```
DELL@KHADEER MINGW64 ~/Downloads
```

```
$ ssh -i Ohiokey.pem ec2-user@3.16.213.163
```

```
The authenticity of host '3.16.213.163 (3.16.213.163)' can't be established.
```

```
ED25519 key fingerprint is: SHA256:zMF+YH7ukIqyfVYH12V0YWxp6QkChqzxTyFnRE0CjUY
```

```
This key is not known by any other names.
```

```
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
```

```
Warning: Permanently added '3.16.213.163' (ED25519) to the list of known hosts.
```

```
** WARNING: connection is not using a post-quantum key exchange algorithm.
```

```
** This session may be vulnerable to "store now, decrypt later" attacks.
```

```
** The server may need to be upgraded. See https://openssh.com/pq.html
```

```
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```

```
Amazon Linux 2023
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https://aws.amazon.com/linux/amazon-linux-2023
```

```
[ec2-user@ip-192-0-0-52 ~]$ ping 18.118.112.166
```

```
PING 18.118.112.166 (18.118.112.166) 56(84) bytes of data.
```

TASK 2

Configure VPC Endpoints for Private AWS Service Access

VPC Endpoints allow resources inside a VPC to reach AWS services over the AWS private network backbone, removing the need for an internet gateway or NAT gateway for that traffic. There are two relevant types: Gateway Endpoints (Amazon S3 and DynamoDB only, attached directly to route tables at no hourly cost) and Interface Endpoints (powered by AWS PrivateLink, used for most other AWS services, billed hourly plus data processing).

2.1. Identify Required AWS Services

30. List every AWS service each application VPC calls today (e.g., S3, DynamoDB, Secrets Manager, KMS, CloudWatch Logs, SSM).
31. Classify each service as Gateway-Endpoint-eligible (S3, DynamoDB) or Interface-Endpoint (everything else).
32. Confirm the AWS Region supports the endpoint type for each required service (most do, but confirm for newer services).

NOTE: Gateway Endpoints have no hourly charge and no data-processing charge, making them the preferred option whenever the service supports it (currently S3 and DynamoDB only).

2.2. Create Gateway Endpoints (S3 and DynamoDB)

33. In the VPC console, go to Endpoints > Create Endpoint.
34. Choose AWS services as the service category, then search for `com.amazonaws.<us-east-2>.s3`.
35. Select the Gateway type (not Interface) when both are listed for S3.
36. Choose the target VPC, then select every route table that should receive the S3 prefix-list route automatically.
37. Optionally attach an endpoint policy to restrict access to specific buckets or actions.
38. Choose Create endpoint, then repeat the same steps for `com.amazonaws.< us-east-2>.s3`.



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Create endpoint Info

Create the type of VPC endpoint that supports the service, service network or resource to which you want to connect.

Endpoint settings

Specify a name and select the type of endpoint.

Name tag - optional

Creates a tag with a key of 'Name' and a value that you specify. Tags help you find and manage your endpoint.

my-endpoint-s3

Type Info

Select a category

- ☒ **AWS services**
Connect to services provided by Amazon with an Interface endpoint, or a Gateway endpoint
- ☐ **PrivateLink Ready partner services**
Connect to SaaS services which have AWS Service Ready designation with an Interface endpoint. Uses AWS PrivateLink
- ☐ **AWS Marketplace services**
Connect to SaaS services that you have purchased through AWS Marketplace with an Interface Endpoint
- ☐ **EC2 Instance Connect Endpoint**
An elastic network interface that allows you to connect to resources in a private subnet
- ☐ **Resources**
Connect to resources like Amazon Relational Database Services (RDS) with a Resource endpoint. Uses AWS PrivateLink
- ☐ **Service networks**
Connect to VPC Lattice service networks with a Service network endpoint. Uses AWS PrivateLink
- ☐ **Endpoint services that use NLBs and GWLBs**
Find services shared with you by service name. Connect to a Network LoadBalancer (NLB) service with an Interface endpoint or to a Gateway LoadBalancer (GWLB) service with a Gateway Load Balancer endpoint

Service Region

☐ **Enable Cross Region endpoint** Info
Connect to cross Region enabled services.

Network settings

Select the VPC in which to create the endpoint

VPC

Create the VPC endpoint in the VPC in the same AWS Region from which you will access a resource.

vpc-0ee7f2fbace051cfb (pub-vpc)

Additional settings

Route tables (1/4) Info

Search						
☐	Name	Route Table ID	Main	Associated Id		
☐	-	rtb-050300b07b5cf50ec	No	-		
☐	-	rtb-018e529591fc54e38	Yes	-		
☐	public-rt	rtb-0da12e4d28ea30494 (public-rt)	No	subnet-0d3991549479859ef (public-subnet)		
☒	private-rt	rtb-00b92f8ee2bce4f23 (private-rt)	No	subnet-065bd923d58c59c2a (private-subnet)		

ⓘ When you use an endpoint, the source IP addresses from your instances in your affected subnets for accessing the AWS service in the same region will be private IP addresses, not public IP addresses. Existing connections from your affected subnets to the AWS service that use public IP addresses may be dropped. Ensure that you don't have critical tasks running when you create or modify an endpoint.

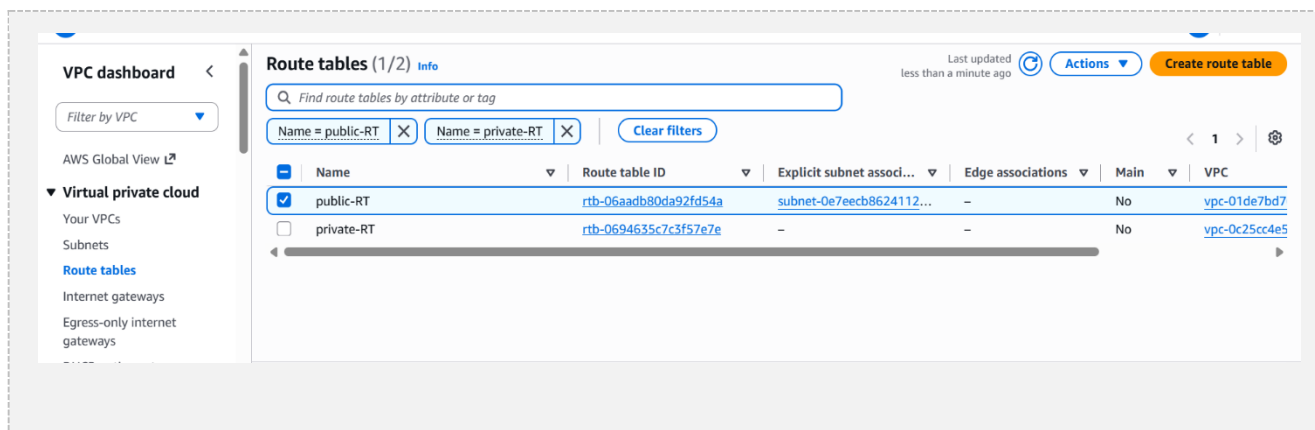
rtb-00b92f8ee2bce4f23 ✕

2.3. Verify Route Tables and DNS Resolution

39. For Gateway Endpoints, confirm the associated route tables now show a route to the pl-xxxxxxx prefix list with target vpce-xxxxxxx.
40. For Interface Endpoints, confirm Enable DNS hostnames and Enable DNS support are both turned on for the VPC (VPC console > Your VPCs > Actions > Edit VPC settings).
41. From a test EC2 instance with no internet gateway or NAT route, resolve the service hostname and confirm it returns a private (10.x/172.x/192.168.x) IP address.



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2.4. Test Private Service Access End-to-End

42. From a private-subnet EC2 instance with no NAT gateway or internet gateway route, call an S3 API action (e.g., list objects in a bucket).
43. Call an Interface-Endpoint-backed service, e.g., retrieve a secret from Secrets Manager, using the standard AWS CLI or SDK with no custom endpoint URL.
44. Confirm both calls succeed and that VPC Flow Logs show no traffic leaving through an internet gateway or NAT gateway for these calls.

